

2(a)(i)	<i>any 1 from:</i> 1 time of soaking (grain in the salt solutions) / 12 hours soaking for each set / AW ; 2 number of grains (soaked in each, salt concentration / set / covered petri dish) ; 3 temperature (of germination / incubation) ; <i>20 °C if quoted</i> 4 time intervals of recording (germination) ; A recorded at 8 hour intervals or recorded over 5 days 5 <i>idea of</i> taking a standard appearance of grain when judged to be germinated ; e.g. <i>emergence of radicle</i>	1
2(a)(ii)	<i>any 1 from:</i> 1 volume of (salt) solution used, on the filter paper / in the (Petri) dish ; I 'amount' unqualified 2 age of grain ; 3 (use) undamaged / not infected / not diseased / AW, grain ; I size / mass, of grain 4 <i>idea of</i> light (exposure of grains during germination / breaking dormancy) ; 5 supply of, air / oxygen (to the grain) ; 6 <i>idea of</i> spacing of grains ; 7 pH (of solution) ;	1

Question	Answer					Marks
2(b)(i)		concentration of salt solution /mmoldm ⁻³	mean (cumulative) percentage of grains that had germinated each day			
			1		5	
			X	Y	X	Y
		0	93.8	92.0	98.0	97.0
		10	93.0	90.2	97.4	96.4
		20	92.2	87.0 ;	96.6	93.6
		30	96.4 ;	90.0	96.4	93.0
		40	92.0	90.4	95.4	92.4
		50	91.3	91.0	95.0	91.6
		60	91.2	90.0	95.0	91.0
2(b)(ii)	1 Pearson's linear correlation ; A Pearson's R Pearson's Rank correlation					2
	2 data collected is continuous					
	or data, is / seems to be, (from a population that is) normally distributed					
	or data / results, appears to be linear ;					

Question	Answer	Marks
2(c)	<i>I data quotes unqualified</i> any 3 from: 1 <i>idea that</i> (germination is rapid because) nearly all / approx. 90% / majority / most, of grains, germinated during the first day ; 2 <i>idea that</i> (control shows) some of the barley, will not germinate, over the period of the experiment / in the first five days ; 3 percentage germination (generally), decreases as concentration of salt increases / increases as concentration of salt decreases ; A <i>idea of</i> <u>negative</u> correlation 4 more grains have germinated after 5 days (than after 1 day) / ora ;	3
2(d)(i)	<i>must be comparative</i> <i>I data quotes unqualified</i> germination of barley is, higher / highest, in X (than Y at, all salt concentrations / every value / stated value(s) from 30 mmol dm⁻³ to 60 mmol dm⁻³) ;	1
2(d)(ii)	any 1 from: 1 <i>idea of</i> measuring / recording / investigating / AW, germination of, X and Y / both, in salty <u>soil</u> ; 2 <i>idea of</i> measuring / recording / investigating / AW, growth of, X and Y / both, in salty <u>soil</u> ; 3 <i>idea of</i> measuring / recording / investigating / AW, yield of, X and Y / both, in salty <u>soil</u> ; 4 AVP ; e.g. a field investigation involving a transect across an area from low to high salt then measuring abundance of the types along the transect	1